

Effect of Entrapped Non-condensable Gas on Shock-Driven Droplet Deformation: Implications for Hypersonic Rain Environments

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Motivation – Rain Erosion

- Hypersonic glide vehicles encounter atmospheric hydrometeors during terminal descent at 0–7 km
 - Micron cloud droplets to millimeter raindrops
- Cumulative droplet impacts roughen the surface → early boundary-layer transition → heating/drag/control-authority penalties
- These margins are already thin in air-breathing systems

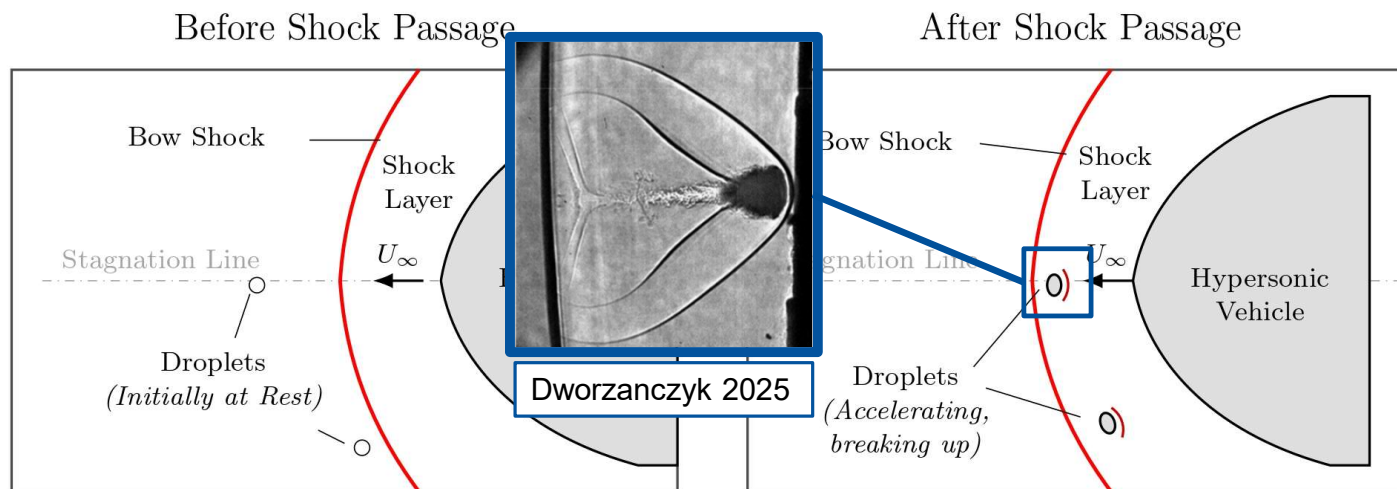


Figure: Schematic of atmospheric droplets interacting with the bow shock of a hypersonic vehicle before and after shock passage. Adapted from Aguilera 2023.

Motivation – Rain Erosion

- Impact loading scales with $\sim \rho D^2 V^2$
- A single 2mm raindrop can exert up to 40kN of force

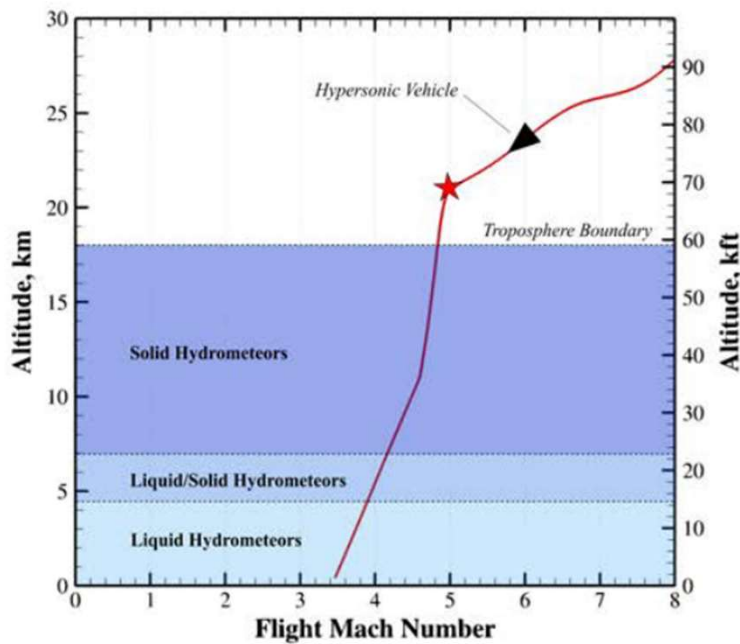


Figure: View of hypersonic vehicle terminal dive phase (Aguilera 2023).



Figure: 4mm droplet impact on radome at Mach 3 (Moylan 2013).

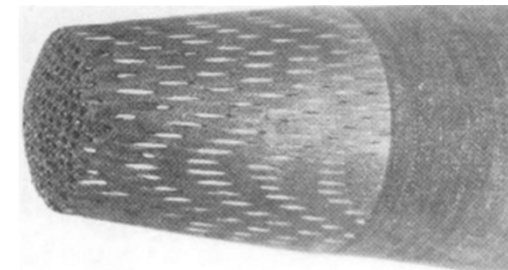


Figure: Vehicle nosetip post flight through clouds (Sandia National Labs).

Motivation – Drop Acoustics and Rain Formation

- Shock transmission → internal wave dynamics
 - Reflections / expansion waves
- Cloud droplets and raindrops form through condensation on suspended aerosol particles
 - May trap microscopic gas pockets (heterogenous nuclei)

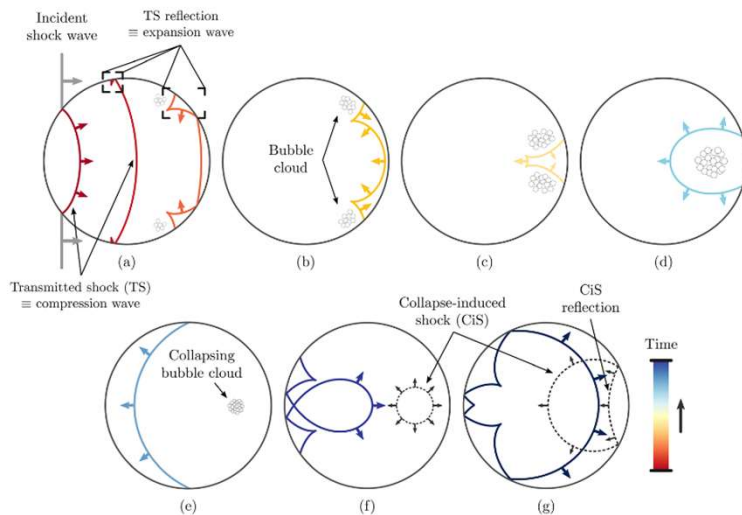


Figure: Progression of droplet internal waves (Biasiori-Poulanges 2023).

Particle	Diameter, μm	Terminal velocity, m s^{-1}
Typical condensation nucleus	0.2	1.0×10^{-6}
Typical cloud drop	20	0.01
Cloud drop / raindrop conventional borderline	100	0.27
Typical raindrop	2000	6.50

Table: Typical sizes and terminal velocities of liquid particles in clouds (Aguilera 2023).



Scope and Objectives

Can the consideration of pre-existing nuclei measurably affect droplet deformation?

Computationally study the role of pre-existing **non-condensable gas nuclei** in the **early-time** response of a shock-loaded liquid droplet.

Objectives:

1. Benchmark the compressible multiphase framework
2. Characterize the “clean” shock-droplet baseline
3. Quantify the mechanical response of embedded gas nuclei
4. Assess the impact of the bubble response on the droplet pressure field and deformation

Literature: Cavitation and Entrained Nuclei

Internal pressure focusing can activate cavitation or gas nuclei, but existing studies have not isolated the mechanical response of pre-existing non-condensable nuclei.

Sembian et al., 2016

- Provided direct shock-water-column evidence of internal wave focusing and cavitation

Forehand et al., 2023-2024

- Compared simulations with and without cavitation modeling
- One-way coupling bubble dynamics

Briggs / Aguilera, 2023-2026

- Experiments of degassed, ambient, and doped droplets ($\sim$ Mach 3, $D=1-3\text{mm}$)
- Limited diagnostic, little evidence

Literature: Knowledge Gaps

Prior work identified cavitation-prone regions and droplet breakup mechanisms but did not resolve the direct feedback of pre-existing gas nuclei.

Gap 1:

Isolated nuclei response

Gap 2:

Two-way coupled multi-nuclei effects

Gap 3:

Early pressure field feedback

Methodology: Computational Setup

All simulations are performed in cylindrical coordinates (x, r) under an axisymmetric assumption.

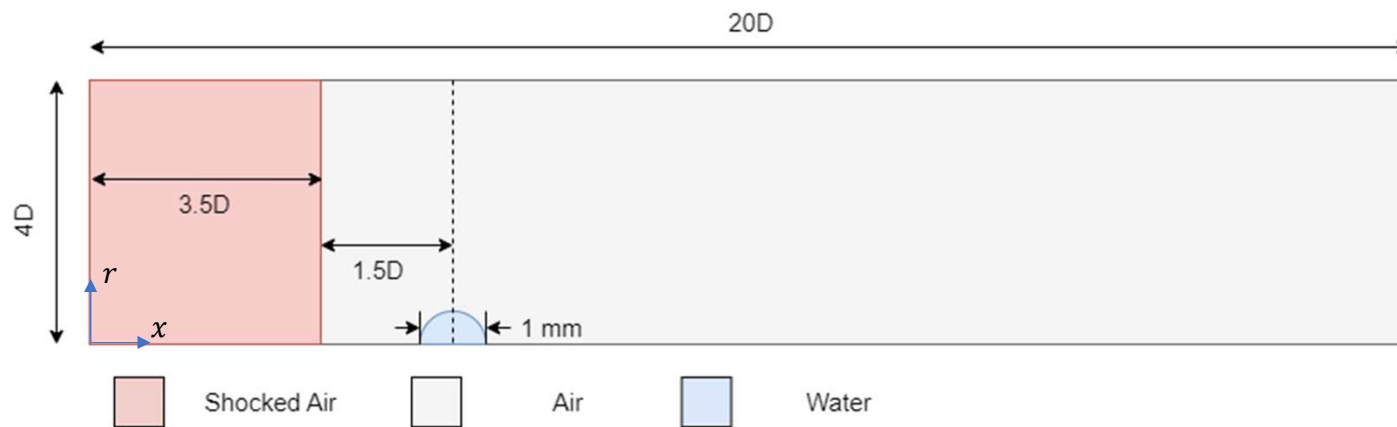


Figure: Computational domain, normalized by droplet diameter, D .

Boundary conditions:

- The bottom boundary is an axis of symmetry
- All other boundaries are non-reflecting boundaries

Initial conditions:

- Shocked air set by Rankine-Hugoniot jump conditions into quiescent, ambient air

Methodology: Simulation Matrix

Parameter space: For a 1mm droplet, $M_s \in \{3, 5, 7\}$, $R_0 \in \{0.1, 0.2, 1, 2\} \mu m$, $N_b \in \{10, 100, 1000\}$

- A clean case is run at each Mach, 36 doped cases

Parameter	Values			
Shock Mach, M_s	3	5	7	-
Initial bubble radius, R_0 (μm)	0.1	0.2	1	2
Number of bubbles, N_b	10	100	1000	-

Table: Shock-droplet simulation matrix. This spans the doped cases.

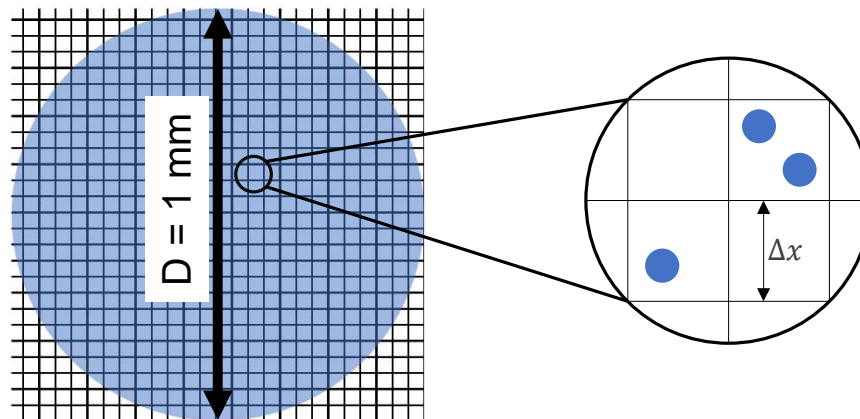


Figure: Sub-grid Lagrangian bubble representation.
 ● : Sub-grid Lagrangian bubble

Methodology: Number Density and Void Fraction

Number density:

$$n = \frac{N_b}{V_{drop}}, \quad V_{drop} = \frac{4}{3}\pi \left(\frac{D}{2}\right)^3$$

- For $N_b = 10, 100, 1000$, n yields $19, 191, 1910 \text{ mm}^{-3}$, respectively

Void Fraction:

$$\beta_0 = \frac{N_b \frac{4}{3}\pi R_0^3}{V_{drop}}$$

	R_0			
N_b	$0.1 \mu m$	$0.2 \mu m$	$1 \mu m$	$2 \mu m$
10	8×10^{-11}	6.4×10^{-10}	8×10^{-8}	6.4×10^{-7}
100	8×10^{-10}	6.4×10^{-9}	8×10^{-7}	6.4×10^{-6}
1000	8×10^{-9}	6.4×10^{-8}	8×10^{-6}	6.4×10^{-5}

Table: Initial global gas void fraction β_0 for the bubble-loading parameter sweep.

Methodology: The Numerical Model

The present simulations are performed in the Multi-Component Flow Code (MFC)

- Open-source, high-order, compressible multiphase solver
- Sub-grid bubble models
- Highly parallel

Resolved Eulerian phase interface:

- Reduced five-equation diffuse-interface formulation of Allaire et al. 2002.

$$\frac{\partial(\alpha_i \rho_i)}{\partial t} + \nabla \cdot (\alpha_i \rho_i \mathbf{u}) = 0, \quad i = 1, 2, \dots, N$$

$$\frac{\partial(\rho \mathbf{u})}{\partial t} + \nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u} + p \mathbf{I}) = 0$$

$$\frac{\partial(\rho E)}{\partial t} + \nabla \cdot [(\rho E + p) \mathbf{u}] = 0$$

$$\frac{\partial \alpha_i}{\partial t} + \mathbf{u} \cdot \nabla \alpha_i = K \nabla \cdot \mathbf{u}$$

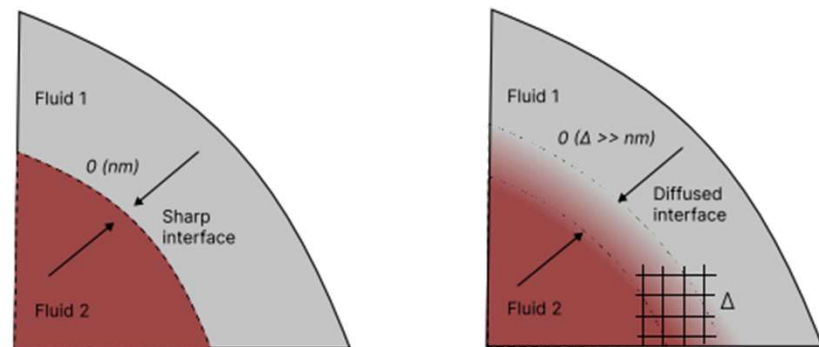


Figure: Sharp and diffuse interface of a multiphase flow (Adebayo 2025).

Methodology: Interface Formulation

Allaire's Model ($\frac{\partial \alpha_i}{\partial t} + u \cdot \nabla \alpha_i = 0$)

- Mechanical equilibrium with an isobaric closure
- Passively advected volume fraction
 - High numerical robustness
- Strictly conservative volume-fraction
- Strict entropy consistency is not guaranteed

Kapila's Formulation ($\frac{\partial \alpha_i}{\partial t} + u \cdot \nabla \alpha_i = K \nabla \cdot \mathbf{u}$)

- Thermodynamically consistent treatment in transition zones
 - However, the smeared region is not physically a transition zone
 - Unclear whether this “fix” is physical
- Needs additional conservative treatment
- Wood's speed of sound is non-monotonic

Methodology: The Numerical Model

Each material is closed with a stiffened-gas equation of state

$$p_i = (\gamma_i - 1)\rho_i e_i - \gamma_i \pi_{\infty,i}$$

The material properties are listed below:

Quantity	Air / bubble gas	Liquid water
Density, ρ	1.17 kg m ⁻³	1000 kg m ⁻³
Ratio of specific heats, γ	1.4	4.4
Stiffened pressure, π_{∞}	0	600 MPa

Table: Equation-of-state and material parameters used in the shock-droplet simulations.

Viscous stresses and surface tension are neglected in the resolved Eulerian equations.

Methodology: Euler-Lagrange Bubble Model

The doped-droplet cases contain N_b Lagrangian gas nuclei embedded inside the initially liquid droplet.

- Each nucleus is modeled as a spherical sub-grid bubble
- The bubbles do not advect
- The resolved Eulerian field provides the local driving pressure
- Bubble volume change feeds back to the Eulerian field through source terms

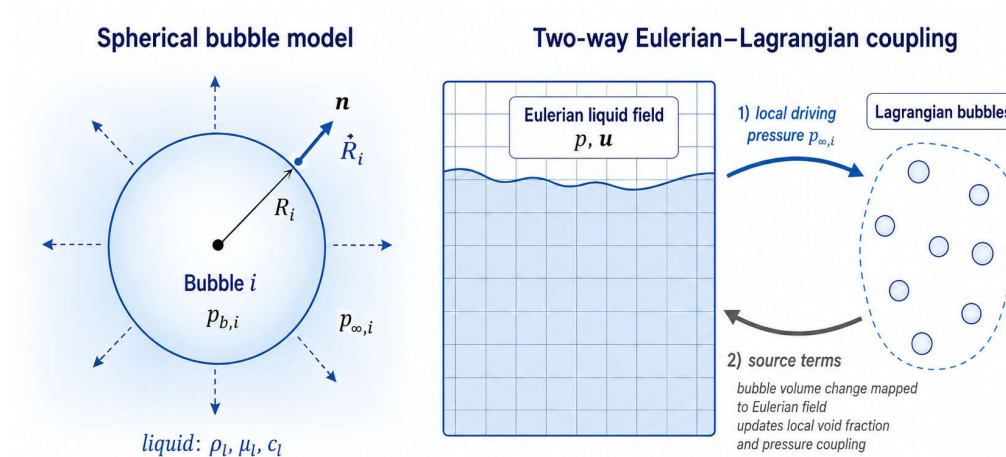


Figure: Sub-grid spherical bubble model and two-way Eulerian-Lagrangian coupling used to evolve the embedded gas nuclei.

Methodology: Keller-Miksis-Type Model

The sub-grid bubble dynamics:

- Bubble radius, $R_i(t)$, is evolved with a Keller-Miksis-type equation

$$\begin{aligned}
 & \underbrace{\left[\left(1 - \frac{\dot{R}_i}{c_l} \right) R_i + \frac{4\mu_l}{\rho_l c_l} \right] \ddot{R}_i + \frac{3}{2} \left(1 - \frac{\dot{R}_i}{3c_l} \right) \dot{R}_i^2}_{\text{Liquid inertia + compressibility/viscous correction} \quad \text{Nonlinear radial inertia}} \\
 &= \underbrace{\frac{1}{\rho_l} \left(1 + \frac{\dot{R}_i}{c_l} \right) \left(p_{b,i} - p_{\infty,i} - \frac{4\mu_l \dot{R}_i}{R_i} - \frac{2\sigma}{R_i} \right)}_{\text{Net pressure driving, corrected for finite sound speed}} + \underbrace{\frac{R_i}{\rho_l c_l} \left(\dot{p}_{b,i} + \frac{4\mu_l \dot{R}_i^2}{R_i^2} + \frac{2\sigma \dot{R}_i}{R_i^2} \right)}_{\text{Time-varying pressure/ acoustic radiation correction}}
 \end{aligned}$$

- Extends the Rayleigh–Plesset equation (RPE) by retaining first-order liquid compressibility effects
 - As $c_l \rightarrow \infty$, the equation reduced to RPE
- Appropriate for spherical sub-grid nuclei in a weakly compressible liquid

Methodology: Numerical Methods

The Eulerian equations are discretized on a structured grid using a finite-volume formulation.

Numerical Component	Method Used
Face reconstruction	2 nd order MUSCL + van Leer limiter
Interface treatment	THINC interface sharpening
Riemann Solver	HLLC
Time-integration	SSP-RK3 (TVD)
Flow-bubble coupling	Operator splitting

Table: Summary of numerical methods used for the Eulerian flow solver and sub-grid bubble dynamics.

A constant CFL value of 0.1 is used in all simulations.



Figure: Numerical results for a planar Mach onto a gas bubble. With THINC (left) and without THINC (right) are both shown (Shyue-Xiao 2014).

Methodology: Strang Splitting

Motivation:

- Two physics with very different timescales:
 - Fluid: pressure waves, advection, governed by CFL timestep
 - Bubbles: Oscillate much faster (stiff)
- Integrating both together forces a tiny Δt everywhere $\rightarrow$ expensive
- Solution: Treat as separate operators

The Scheme:

For $\partial q / \partial t = -\nabla \cdot F(q) + s(q)$, each time step performs three sub-steps:

1. Source half-step ($t \in [0, \Delta t/2]$)

$$\frac{\partial q'}{\partial t} = s(q'), \quad q'(0) = q^n$$

2. Flux full-step ($t \in [0, \Delta t]$)

$$\frac{\partial q''}{\partial t} = -\nabla \cdot F(q''), \quad q''(0) = q'(\Delta t/2)$$

3. Source half-step ($t \in [0, \Delta t/2]$)

$$\frac{\partial q'''}{\partial t} = s(q'''), \quad q^{n+1}(0) = q'''(\Delta t/2)$$

Key Points:

- Half + full + half achieves 2nd-order splitting accuracy
- Full + full achieves 1st-order
- Steps 1 & 3 subcycle internally with an adaptive embedded RK scheme
 - Takes as many steps as needed to reach exactly $\Delta t/2$
- Step 2 uses the standard TVD-RK fluid update at the CFL Δt

Methodology: Code Extensions

Cylindrical Euler-Lagrange Bubble Model: allows Lagrangian gas nuclei to be used in the axisymmetric formulation used for the simulations

- Azimuthal mapping operator to each 3D bubble:

$$T_{\theta}(\cdot) = \frac{1}{2\pi r} \int_0^{2\pi} (\cdot) r d\theta$$

- Preserves equivalent axisymmetric bubble volume
- Deposits bubble source terms consistently in cylindrical coordinates
- Resulting axisymmetric void fraction field:

$$\beta_{axi}(z, r) \equiv (T_{\theta}\beta)(z, r) = \sum_{n=1}^{N_b} V_n \delta_{axi}(z, r; z_n, r_n)$$

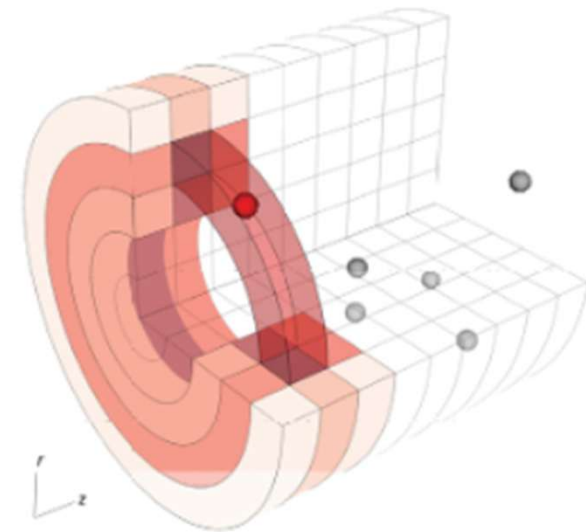


Figure: Axisymmetric mapping of a 3D Lagrangian bubble kernel onto neighboring FVM cells (Maeda-Colonius 2018).

Methodology: Benchmarking

Benchmarking Against Shock-Water Column Experiments:

- Sembian 2016 studies the interaction of a planar shock (Mach = 1.75 and 2.4) onto a droplet column of diameter $D_0 = 22 \text{ mm}$
- Large droplet allows for the observation of key internal acoustics
- Numerical schlieren is compared directly to the experimental shadowgraph
 - Comparison of transmitted shocks, reflected waves, expansion regions, slip surfaces, and triple-point/Mach-stem structures
- Excellent agreement is observed

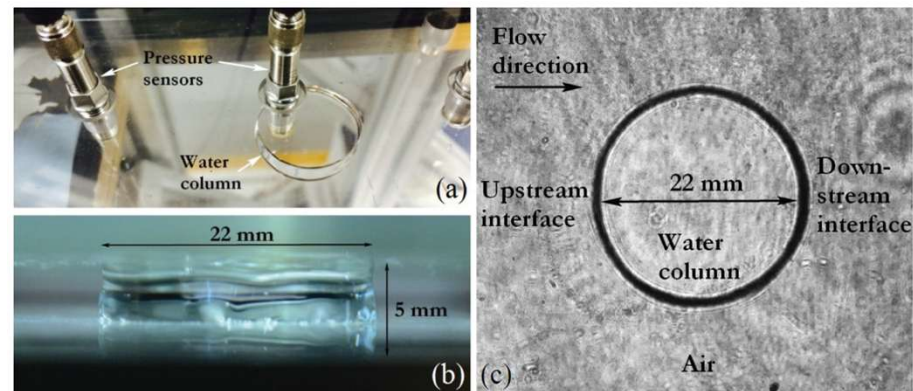


Figure: 22mm water column of Sembian 2016.

Methodology: Benchmarking

Benchmarking Against Shock-Water Column Experiments:

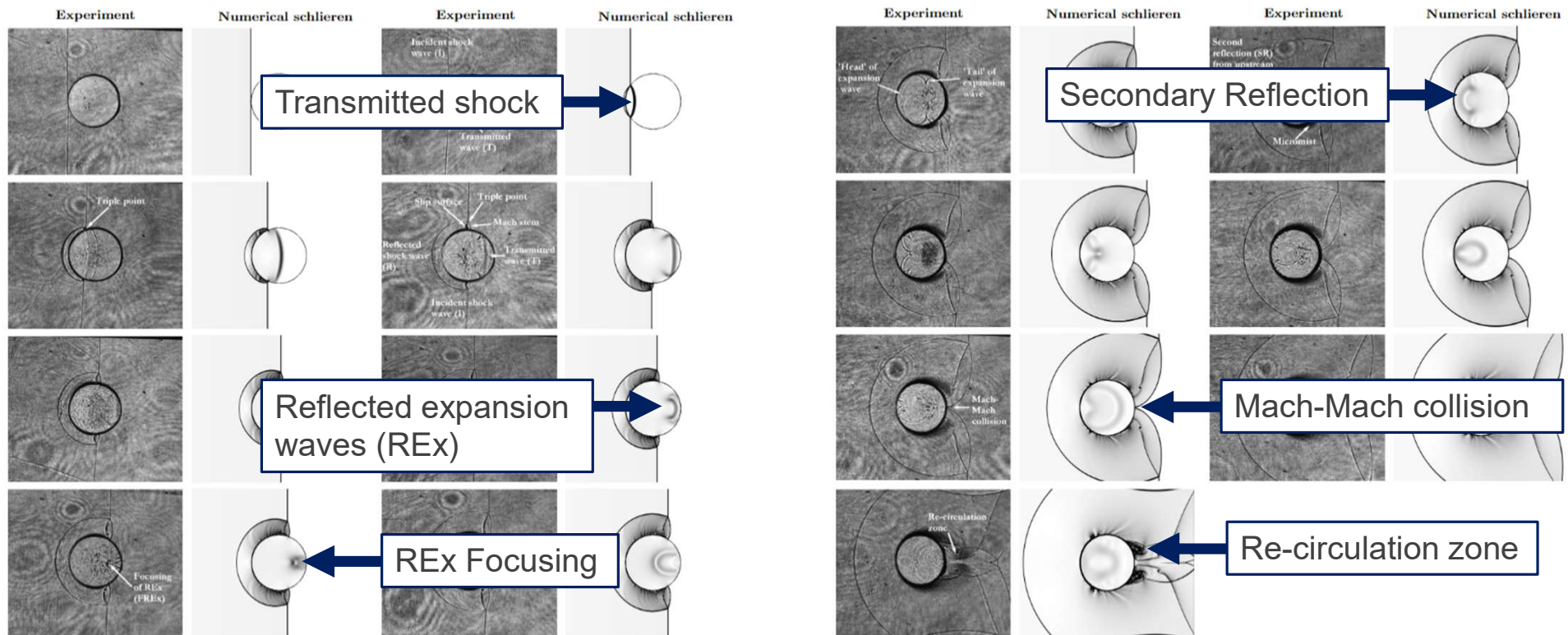


Figure: Comparison between experimental shadowgraph images from the Mach 2.4 shock-water-column experiment of Sembian et al. and numerical schlieren fields from the MFC simulation.

Methodology: Mesh Independence

- Performed on a clean droplet in our case of interest
- Mach 7 is chosen for the study as it produces the largest pressure gradients within the droplet
- A refinement factor of $\sim\sqrt{2}$ is used

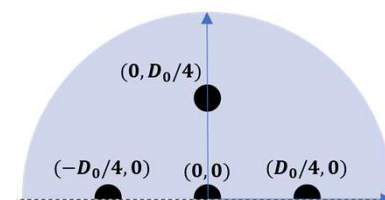
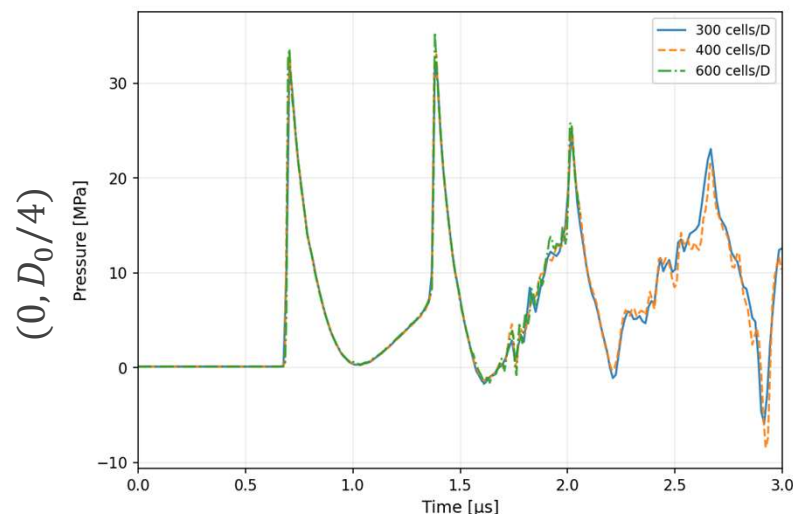
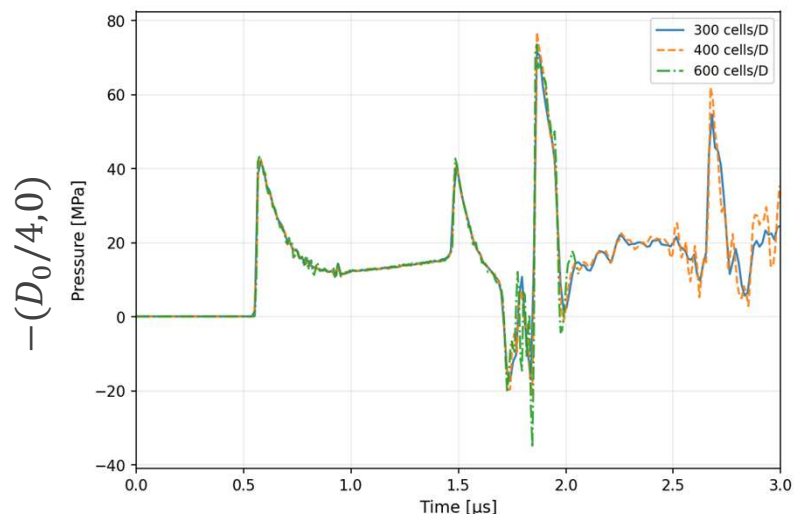
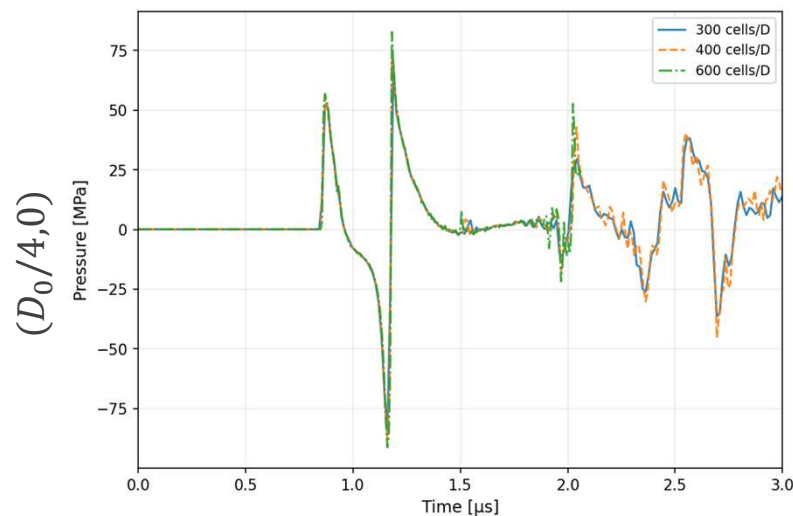
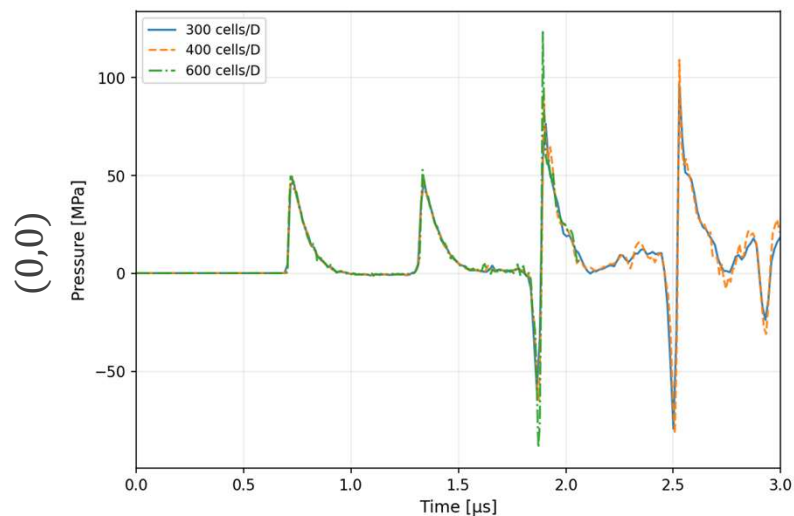
Mesh	Cells / D_0	Grid size ($N_x \times N_r$)	Δx
Coarse	300	6000×1200	$3.33 \mu m$
Medium	400	8000×1600	$2.50 \mu m$
Fine	600	12000×2400	$1.67 \mu m$

Table: Mesh levels used in the shock-droplet mesh-independence study.

- Shock-arrival time, peak pressure, pressure decay, and area-averaged pressure response across mesh levels was compared
- All meshes show close agreement, hence the medium mesh was chosen

Methodology: Mesh Independence

Nearest-cell pressures



Methodology: Summary of Assumptions

Geometry / interface:

- Axisymmetric geometry
- Diffuse-interface formulation with interface sharpening
- Droplet-scale surface tension neglected

Sub-grid nuclei model:

- Bubbles are modeled as spherical sub-grid nuclei
- Bubbles remain fixed in space; no stochastic correction
- No asymmetrical collapse
- Bubble-bubble interaction only through the resolved pressure field

Thermodynamic / transport simplifications

- Non-condensable gas nuclei
- No vapor production, mass transfer, and phase change
- Polytropic bubble gas response
- Neglecting viscosity in Eulerian phase

Clean Results: Non-dimensionalization

Shock-droplet interaction in the absence of dissolved gas nuclei.

- All quantities are presented in the dimensionless time:

$$\tau = \frac{t - t_{arr}}{t_b}$$

- Where t_b is the Ranger-Nichols aerobreakup timescale:

$$t_b = \frac{D}{U} \sqrt{\frac{\rho_l}{\rho_g}}$$

- Define acoustic crossing time:

$$t_c = D/c_w$$

$$\tau_c = t_c/t_b$$

Quantity	$M_s = 3$	$M_s = 5$	$M_s = 7$
$p_{post} [MPa]$	1.044	2.929	5.757
$u_{post} [m s^{-1}]$	772.5	1390.6	1986.5
$t_b [\mu s]$	19.27	9.40	6.31
$t_c = D/c_w [\mu s]$	0.615	0.614	0.613
$\tau_c = t_c/t_b$	0.0319	0.0653	0.0971

Table: Key acoustic timescales.

Clean Results: Internal Wave Structure

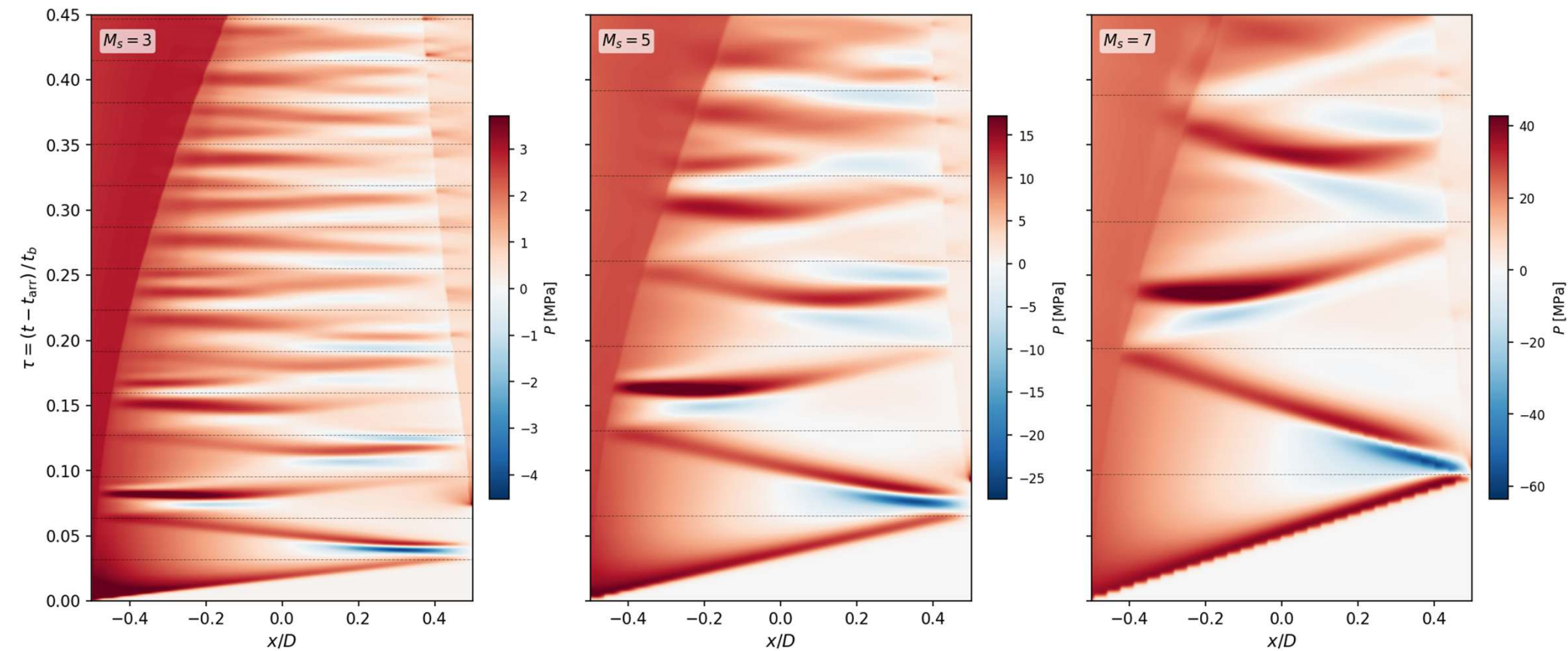


Figure: Space-time diagram of axial centerline pressure for the three clean-baseline cases. Left: Mach 3; center: Mach 5; right: Mach 7.

Clean Results: Peak Pressure

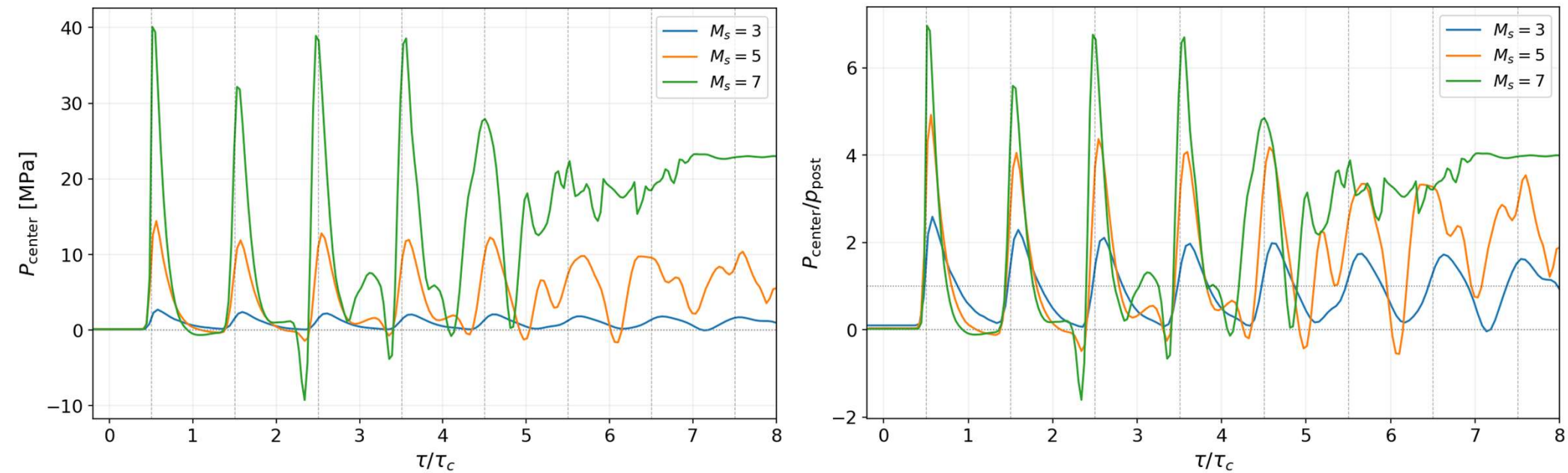


Figure: Center-probe pressure history plotted against time normalized by the acoustic crossing time. *Left:* absolute pressure. *Right:* pressure normalized by case-specific post-shock pressure.

Clean Results: Rarefaction

Each acoustic reflection generates a rarefaction wave that propagates back into the droplet and produces tensile pressures.

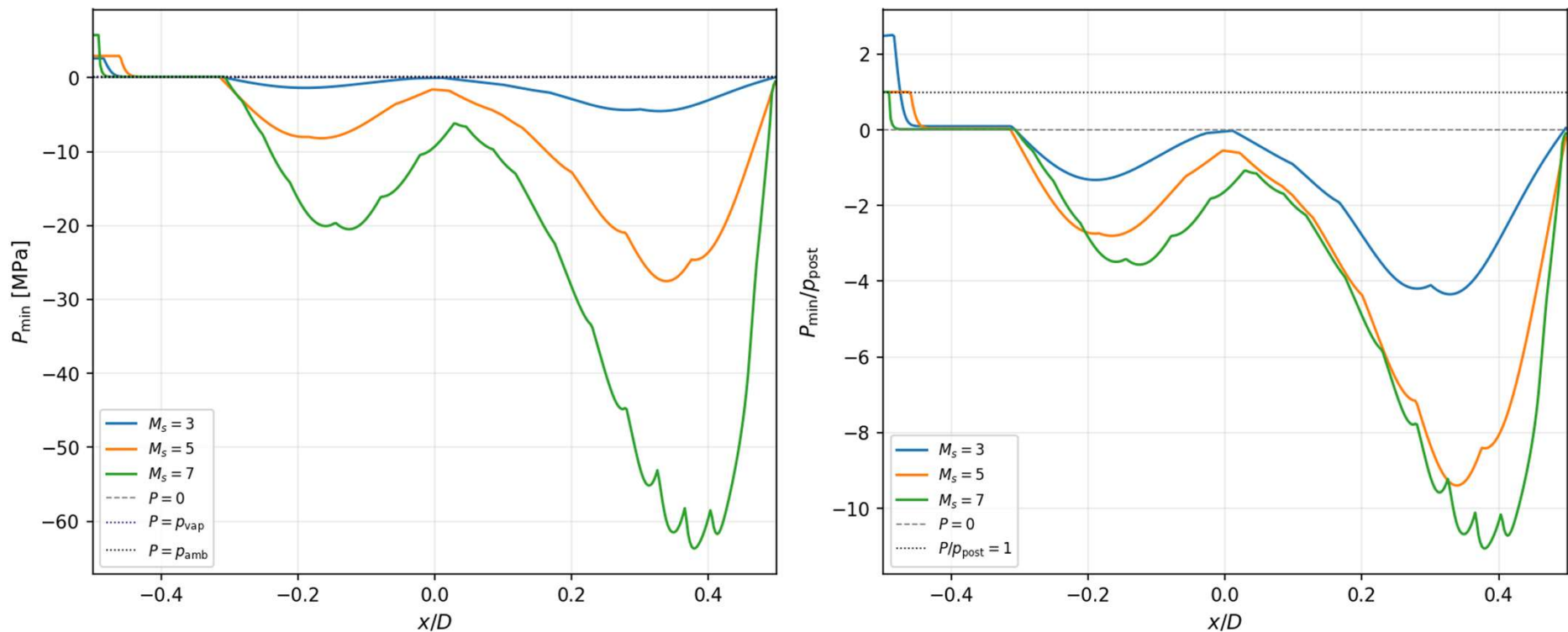


Figure: Minimum centerline pressure along the droplet axis, minimized over $\tau \in [0, 0.45]$. *Left*: absolute minimum pressure. *Right*: normalized by p_{post} .

Clean Results: Early Droplet Deformation

Early-time droplet deformation:

- Noticeable flattening on the windward face
- Rim roll-up and lip formation
- RTP mode formation
- Dimple forming at the aft interface

Numerical artifact*:

- Upstream-face dimple

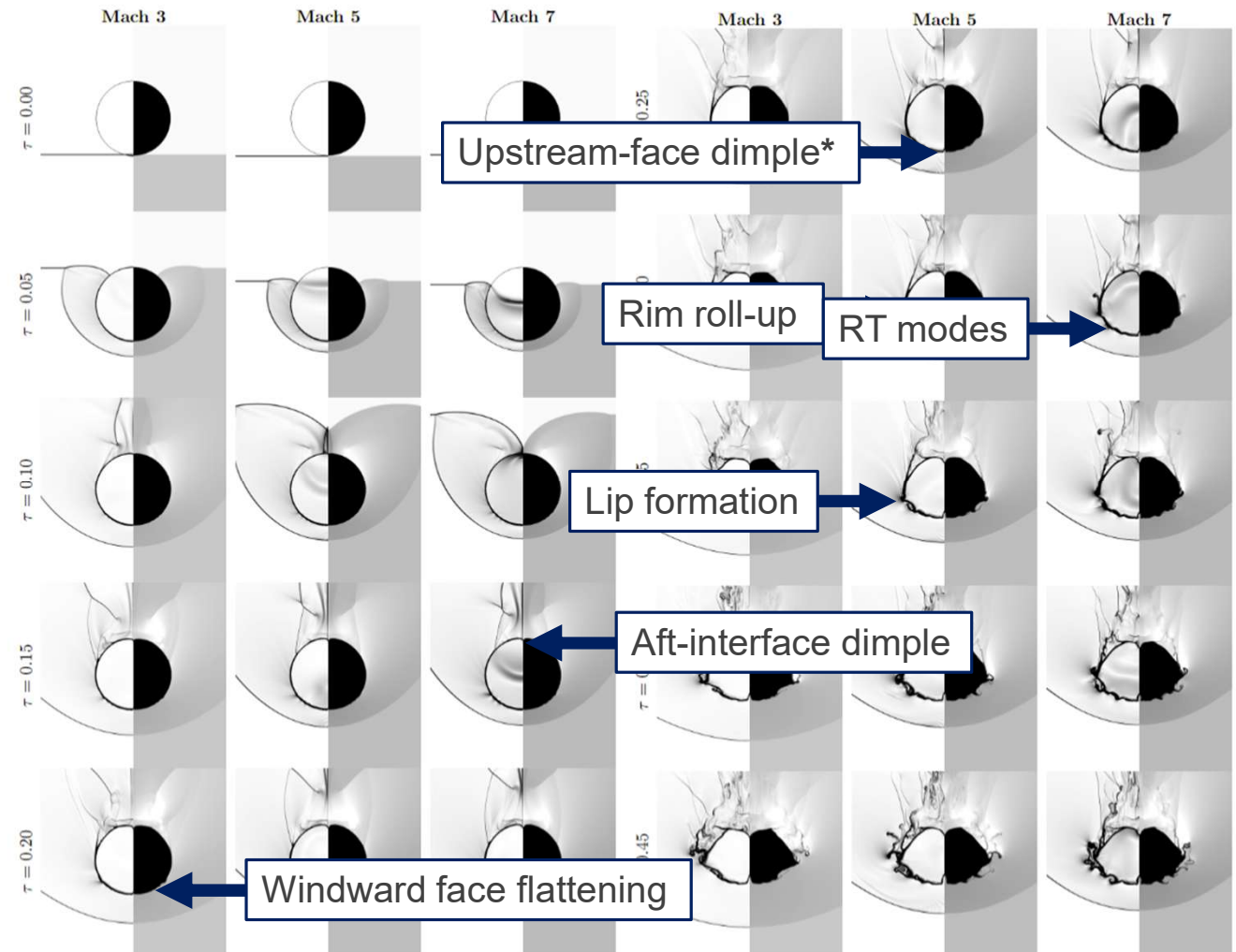


Figure: Numerical schlieren *left* and numerical shadowgraph *right* of the early-time droplet deformation.

Doped Results: Spatial Selectivity of Activation

The normalized underpressure integral predicted that bubble activation should be concentrated in the downstream half of the droplet.

- The simulations confirm this prediction

Consider the Mach 5, $R_0 = 0.2 \mu m$, $N_b = 100$ case:

- Bin the 100 bubbles by initial position

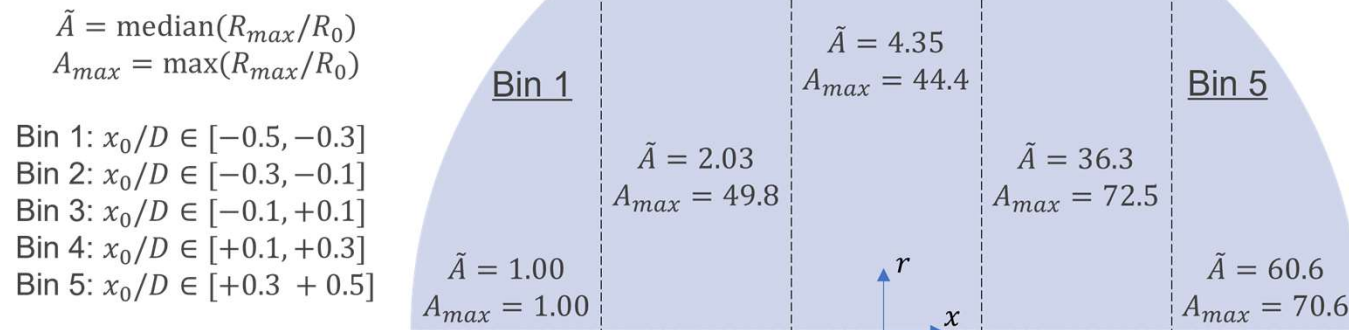


Figure: Spatial bins based on initial streamwise nucleus location, showing median and maximum bubble growth.

Doped Results: Spatial Selectivity of Activation

Downstream half of the droplet shows most bubble activation

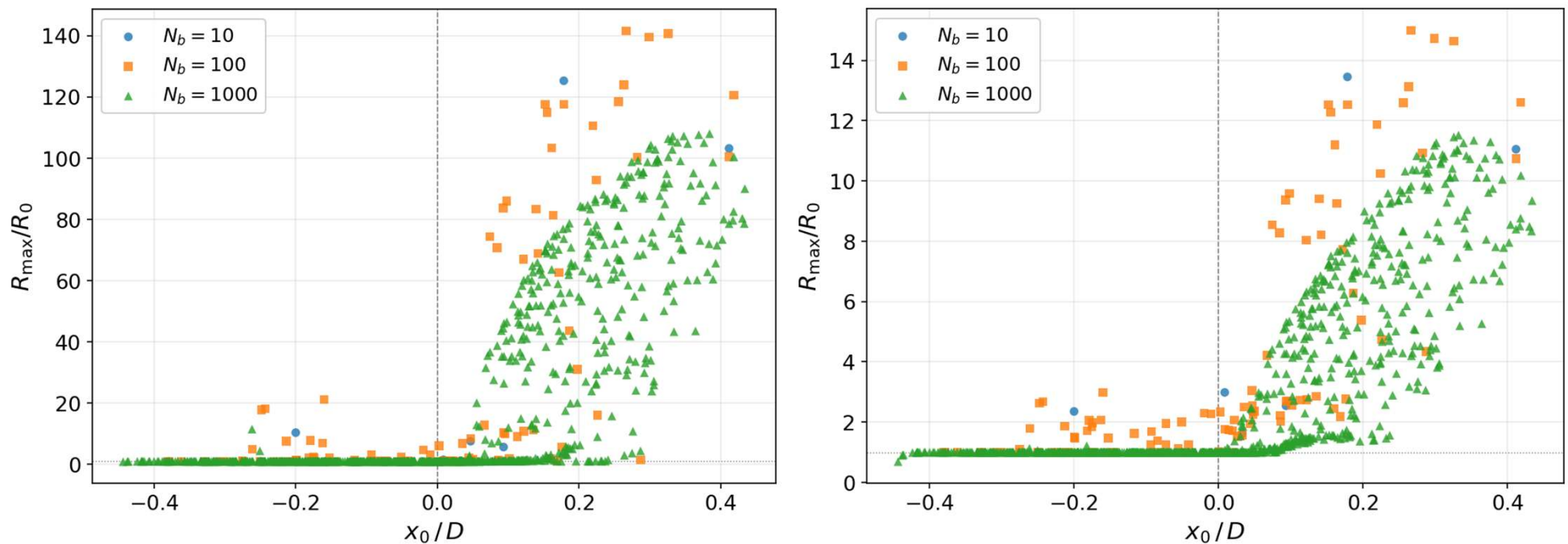
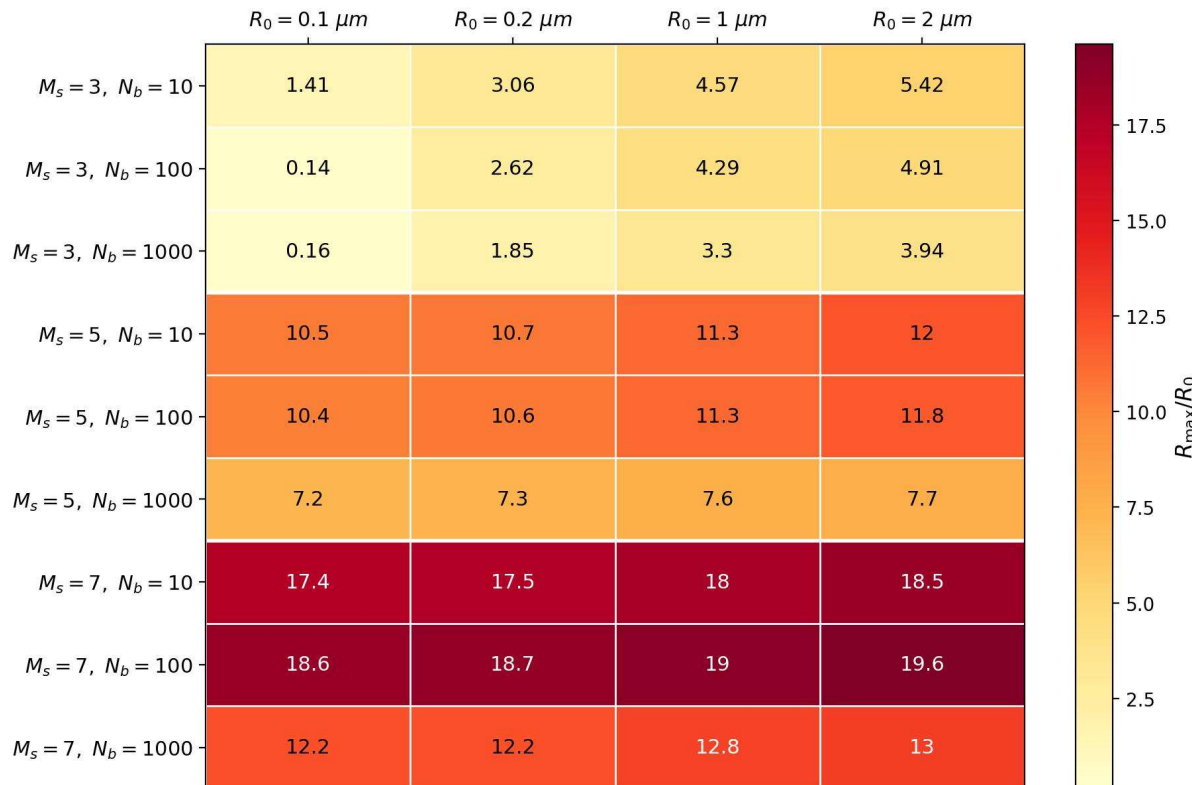


Figure: Scatter plots of normalized bubble growth at Mach 5. Left: $R_0 = 0.1 \mu\text{m}$. Right: $R_0 = 1.0 \mu\text{m}$.

Doped Results: Absolute Bubble Growth



Mach number:

- Strongest influence on peak bubble growth ($Mach \uparrow, R_{max} \uparrow$)

Initial bubble radius:

- Significant effect on R_{max}/R_0
- $\sim$ constant R_{abs} for a given Mach

Bubble population:

- Least effect on peak bubble growth
- $N_b = 1000$ seems to limit peak bubble growth when compared to other population sizes

Figure: Absolute maximum bubble radius in μm for all doped cases.

Doped Results: Pressure Perturbations

The volumetric growth and collapse of the Lagrangian bubbles feeds back onto the Eulerian carrier field

$$P_{doped}(x, \tau) - P_{clean}(x, \tau) = \Delta P(x, \tau)$$

The bubble population attenuates the clean acoustic reverberations

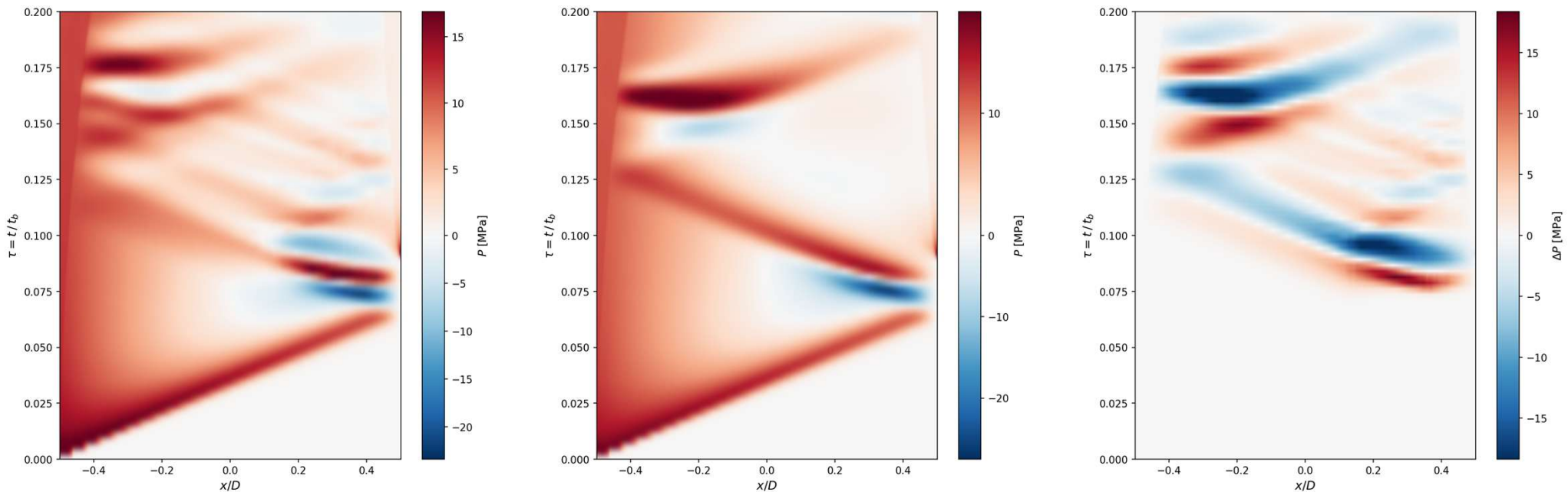


Figure: Space-time diagram of axial centerline pressures at Mach 5. Left: Doped ($R_0 = 0.1\mu m, N_b = 1000$); Center: Clean; Right: pressure difference.

Doped Results: Pressure Perturbations

- The perturbation signal is negligible during the initial shock transmission phase
- The pressure perturbation fills the entire droplet
 - Forms characteristic V-shaped or X-shaped diagonal bands

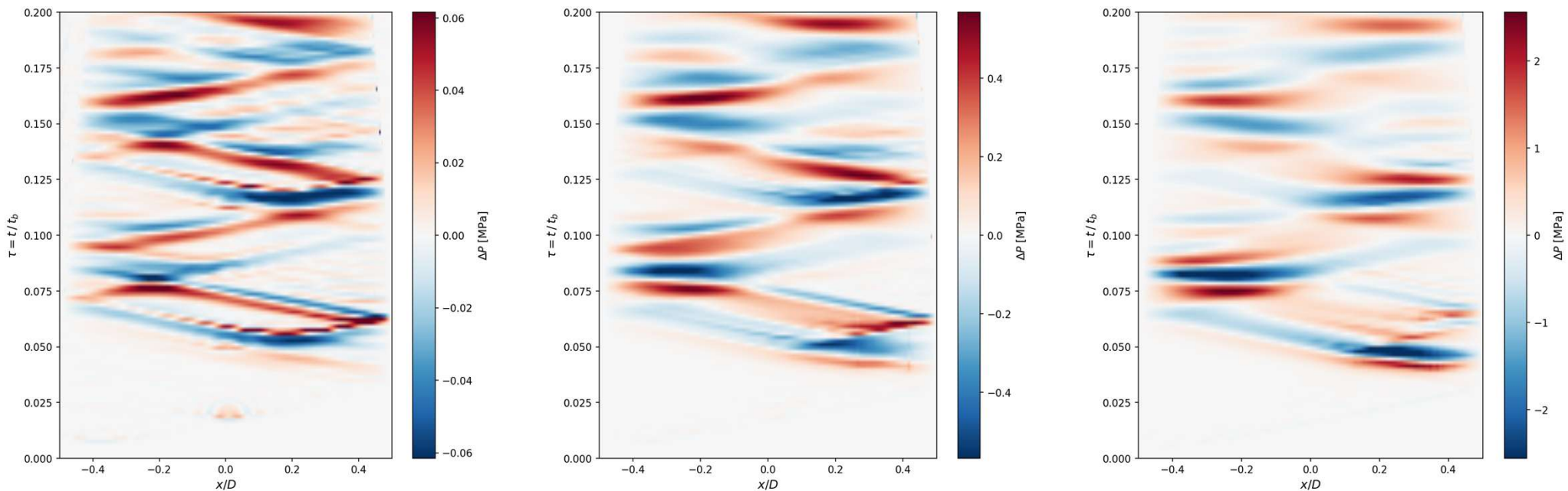
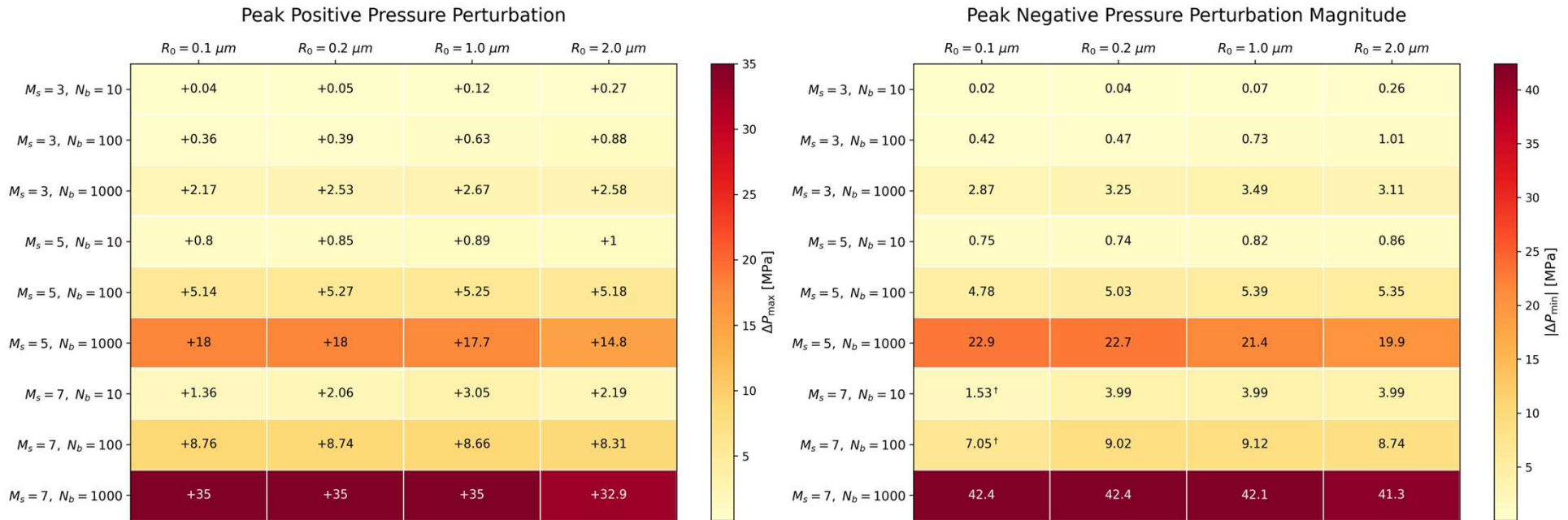


Figure: Space-time diagram of axial centerline pressure differences at Mach 3 at $R_0 = 1.0 \mu\text{m}$.
 Left: $N_b = 10$; center: $N_b = 100$; right: $N_b = 1000$.

Doped Results: Pressure Perturbations



- Peaks are highly dependent on Mach number (Mach $\uparrow$, $|\Delta P_{max/min}| \uparrow$)
- Peaks are highly dependent on bubble population ($N_b \uparrow$, $|\Delta P_{max/min}| \uparrow$)

Doped Results: Early Droplet Deformation

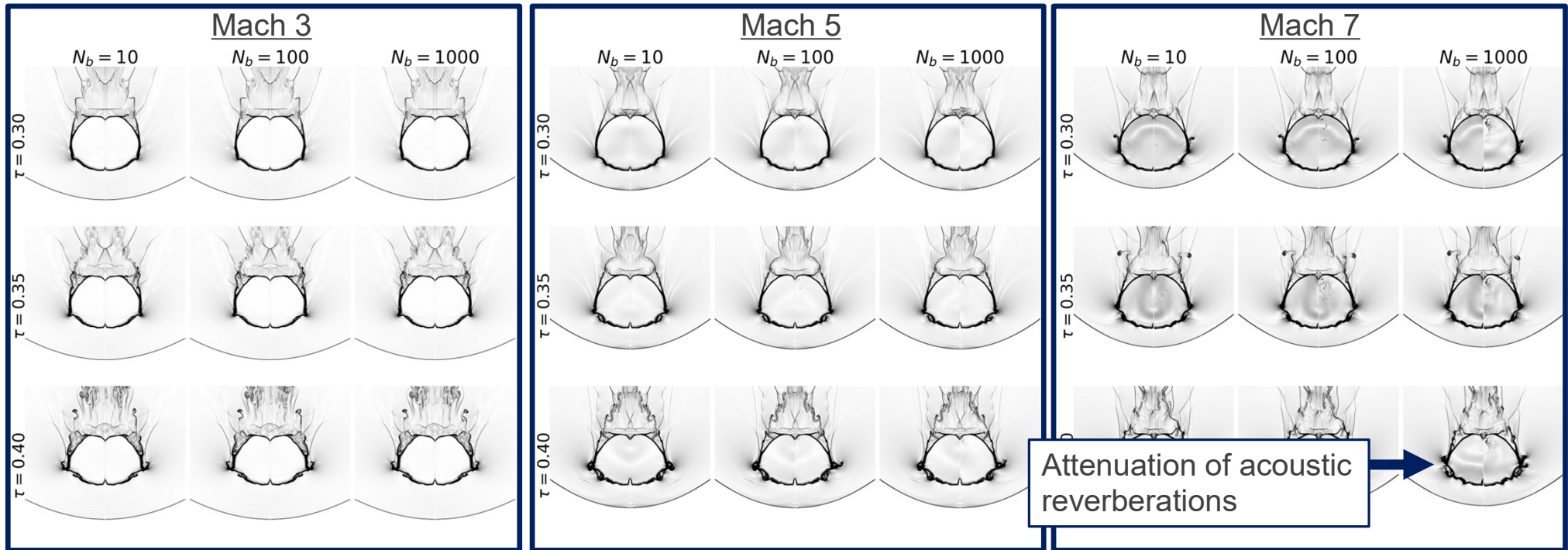
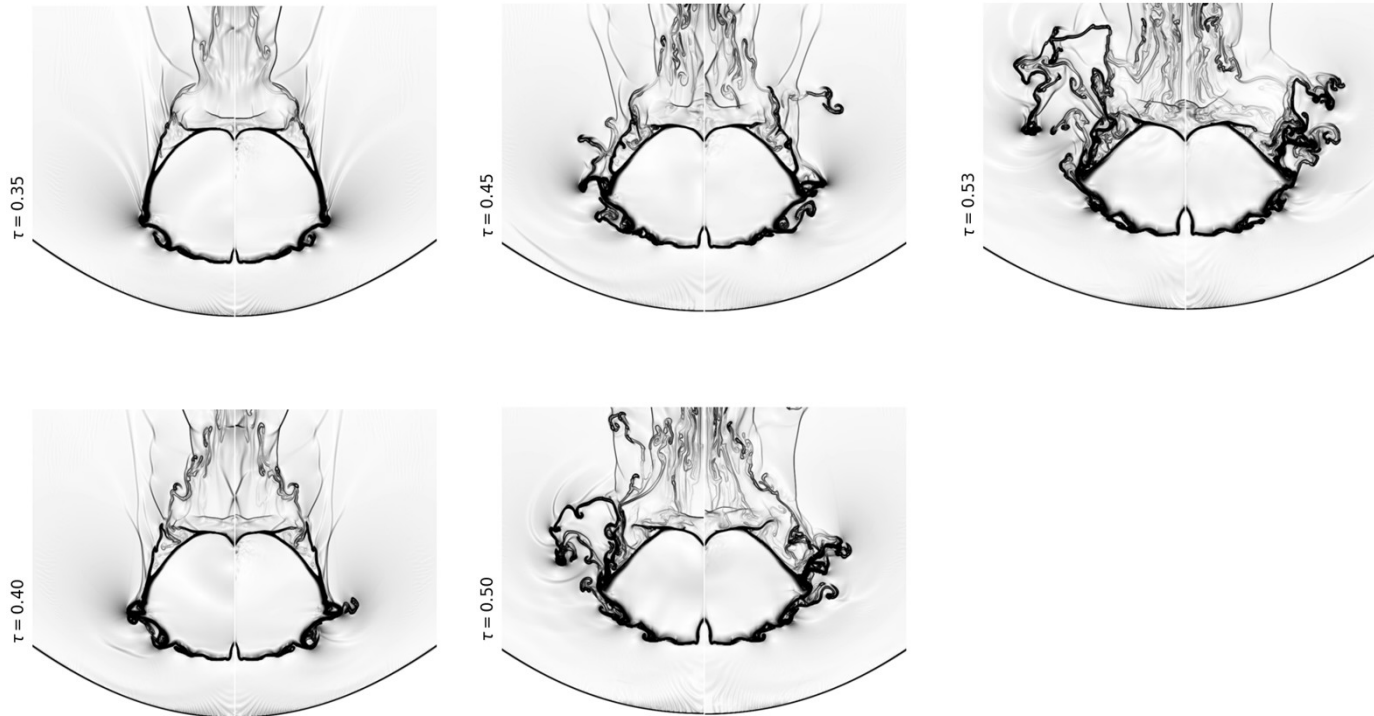


Figure: Numerical schlieren taken at doped cases with $R_0 = 1.0\mu m$. Each frame shows comparison of respective clean half (left) and the doped half (right). Left: Mach 3; center: Mach 5; right: Mach 7.

- There is nearly no observable difference* in surface deformation at any Mach or bubble density
 - *Mach 7, $N_b = 1000$, $\tau = 0.40$
- Noticeable damping of internal acoustics in doped cases with $N_b = 1000$

Doped Results: Early Droplet Deformation

$M_s = 5$, $R_0 = 0.2 \mu\text{m}$, $N_b = 10000$
left: clean | right: doped



Principal Findings

Main Conclusion:

Pre-existing gas nuclei undergo strong internal activation, but this activation does not measurably alter early droplet-scale deformation over the tested parameter range, given the modeling assumptions.

- Clean-droplet cases establish strong internal acoustic focusing
- Gas nuclei are activated primarily by reflected tensile waves
- Bubble growth is spatially selective
- Dense populations produce acoustic attenuation and self-suppression of individual bubble growth
- Despite strong local bubble dynamics and pressure perturbations, early interface deformation remains nearly unchanged

Limitations and Future Work

Limitations:

- Vapor production, condensation, and mass transfer were excluded
- Bubble nuclei were fixed in space and did not advect with the liquid
- Axisymmetric assumption limits 3D instability development
- Conclusions are computational and require targeted experimental validation

Model extensions:

Add rate-limited phase-change models

Allow nuclei to advect with the liquid

Extend simulations beyond early-time axisymmetric regime

Extend nuclei modeling to solid/particulate inclusions and heterogeneous cavitation sites

Experimental direction:

Perform companion experiments with controlled nuclei populations

Acknowledgements

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This work was partially supported by the Air Force Office of Scientific Research
under award number FA9550-22-1-0021.

The author acknowledges DoD HPCMP and ERAU VEGA HPC use.

